



TIPS & TRICKS

A left-to-right storage-saving rsync command for incremental backups

With a small hack, rsync can be used to create tiny incremental backups on file systems supporting hard links, as follows:

```
rsync -avPZH -e ssh --delete --exclude="pattern1.*"
--include="pattern1.pattern2" --link-dest=path_to_yesterdays_
targetdir remote_host:path_to_sourcedir path_to_todays_
targetdir
```

-a means archive and is short for -rptgoD.
-n means dry run. You can remove this after testing.
-v means verbose.
-P means keep partially transferred files and show progress.
-z means use compression to transfer.
-H means preserve hard links.
-X means preserve extended attributes.
-e ssh means use the SSH protocol.
--delete means sync deletions.
--exclude="pattern1.*" --include="pattern1.pattern2"
means exclude all pattern1.* except pattern1.pattern2
--link-dest=path_to_yesterdays_targetdir means hard link
to unchanged files in path_to_yesterdays_targetdir

—A. Datta, webmaster@aucklandwhich.org

Resume broken downloads using wget

We have all had the experience of the network failing just when about 99 per cent of something we're downloading is complete.

This is really annoying when you are downloading a huge file on a rather slow connection. Most browsers support the *Resume* capability, but what if you are using the command line and downloading something using wget?

It's easy; just run the wget command again, but this time, with the -c or --continue parameter.

For example:

```
wget -c https://link/to/file
```

...or:

```
wget --continue https://link/to/file
```

—Jackson Isaac, jacksonisaac2008@gmail.com

Creating an empty file or folder using the cat, xargs, touch and mkdir commands

Here is a simple command to create an empty file. First, list down the file names that you want to create, as follows:

```
$cat > file.txt
hemanth.dat
bitra.dat
bitroid.dat
in.dat
```

Now run the following command:

```
$cat file.txt | xargs -I {} touch /home/username/Desktop/ {}
```

You can check if the empty file has been created by running the following command:

```
$ls /home/username/Desktop/
hemanth.dat
bitra.dat
bitroid.dat
in.dat
```

Similarly, we can create empty folders too, as follows:

```
$cat somefile.txt | xargs -I {} mkdir -p /home/username/
Desktop/ {}
```

...where somefile.txt has the names of folders.

—Hemanth Bitra, hemanthbitra@bitroid.in